String Art

Using the Radon transform and Fourier analysis to reconstruct images using string.

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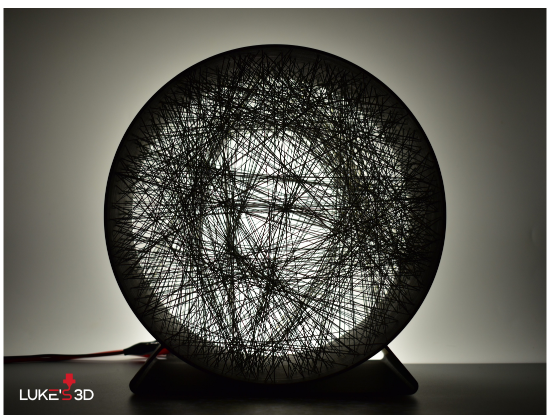
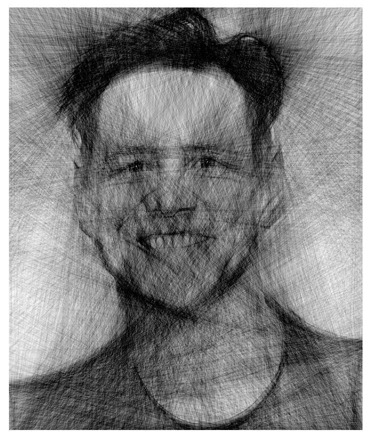
1 Introduction

Art and mathematics share a profound connection, often manifesting in surprising ways. String art, with its intricate patterns and striking visual appeal, represents a fascinating blend of artistic creativity and mathematical rigour.

I first came across string art on a Youtube video that displayed the intricate work of artists. Originally, I had falsely believed the creations were a result of an artistic mind. However, I soon discovered there was no human mind behind the creations: the art was what is called “algorithmic art”. The process is highly mathematical.



(a) The first string art I saw of Einstein.



(b) The final creation of this paper, a portrait of Jimmy Carter.

Despite this, there exists very little work that discusses art in this form because it is developed by commercial artists with no incentive for sharing. Nonetheless, the results at the end of the paper do show a degree of success in the reconstruction.

This paper explores the mathematical foundations underlying the creation of string art, focusing on the use of Radon transforms and Fourier analysis to convert images into patterns of discrete lines. The goal is to mathematically derive and implement a process that transforms an image into a recognizable piece of string art. The first sections of the paper will explore the theorems we will require. When the frameworks are established, we mathematically derive a final solution to creating string art and display our results.

2 The Radon Transform

The Radon transform is a mathematical operation that projects a 2D function $f(x, y)$, such as an image, onto a set of lines at various angles. Originally, the purpose of the transform was to create visualizations of the human body using computed tomography (CT) scans. However, in this paper, the Radon transform can be used to decompose an image into line-based projections that will be useful for string art. The Radon transform projects lines through an image and returns how dark the pixels on the lines are. A metaphor would be shining beams of lasers through an image, and the more opaque areas would lead to a lower value in the Radon transform. The essential idea of string art is that the areas with lower values will be where lines are drawn, although, we will see that it becomes much more complicated than that.

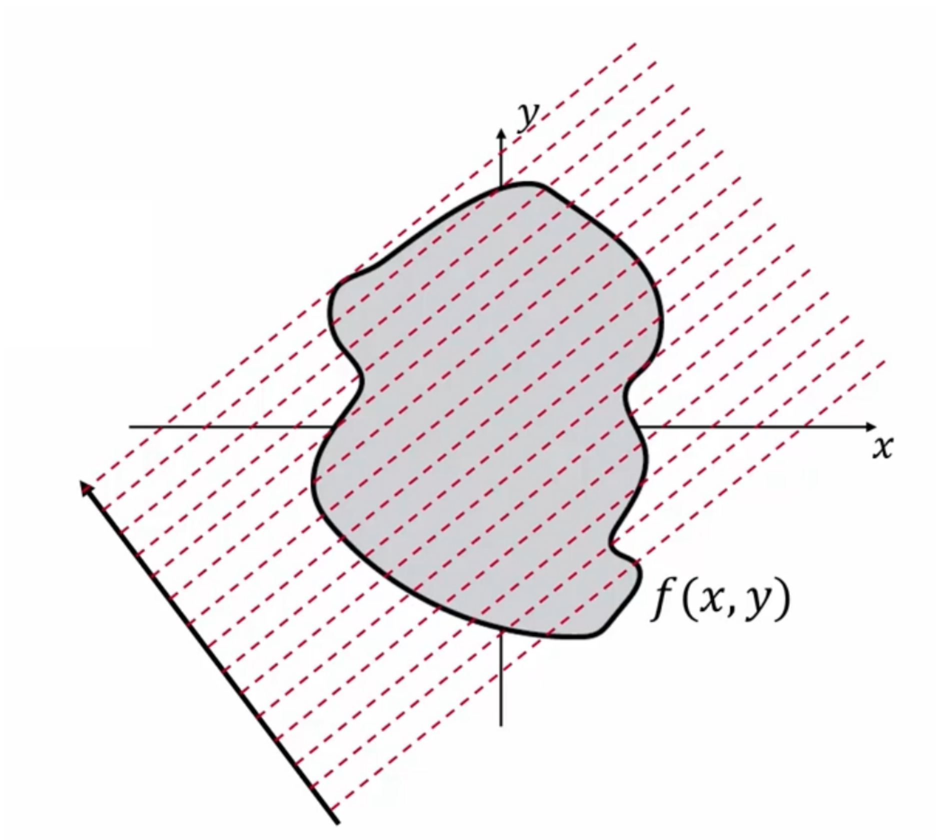


Figure 2: Visual representation of the Radon transform

The Radon transform $\mathcal{R}$ of a function $f(x, y)$ is defined as an integral over lines $L_{s,\phi}$ parameterized by their distance s from the origin and their angle ϕ relative to the x-axis. Mathematically, the Radon transform, $p(s, \phi)$, performed on the object $f(x, y)$ is defined as

$$p(s, \phi) = \int_{L_{s,\phi}} f(x, y)\, du \tag{2.1}$$

where du is a small length along the line $L_{s,\phi}$.

Each line $L_{s,\phi}$ is characterized by the perpendicular distance s from the origin and the angle ϕ, which determines its orientation. To express this line in Cartesian coordinates, we can relate any point (x, y) on $L_{s,\phi}$ to s and ϕ using trigonometric transformations. Specifically:

$$x = s\cos\phi - u\sin\phi \tag{2.2}$$

$$y = s\sin\phi + u\cos\phi \tag{2.3}$$

Where u represents the position along the line in a direction orthogonal to s. Substituting equations 2.2 and 2.3 into equation 2.1, we obtain the final expression for the Radon projection

$$p(s,\phi) = \int_{-\infty}^{\infty} f(s\cos\phi - u\sin\phi, s\sin\phi + u\cos\phi)\, du \tag{2.4}$$

This projection accumulates the values of $f(x, y)$ along each line, yielding a scalar $p(s, \phi)$ that represents the projection of the function f at angle ϕ and distance s. The lower the value, the darker the sum of the pixels was on the line.

2.1 Sinogram Representation

Each iteration of the radon projection results in a scalar value. There are two variables we can iterate through, ϕ and s. When we apply the projection while varying both variables, we can obtain a heatmap that contains the projection over all the possible lines through the image. The result is called a sinogram, shown in Figure 3. The darker the line, the darker the pixel values were on the line drawn across the canvas.

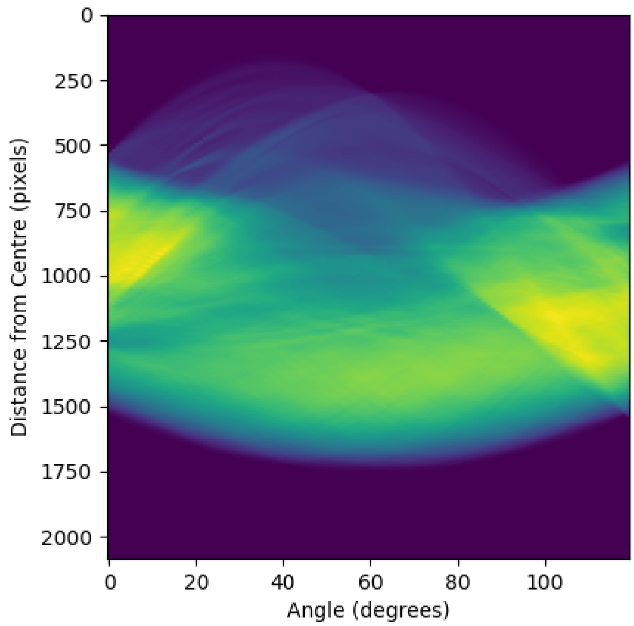
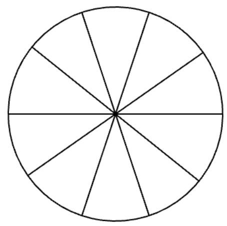


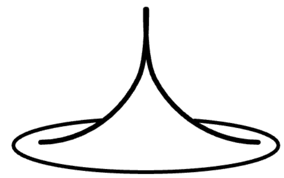
Figure 3: Repeated Radon projection.

3 Back projection

Recall that each pixel corresponds to a line across the canvas. Already, the representation is similar to stretching a string across the canvas. Hence, creating the string art is quite similar to reconstructing the image using the sinogram, since the values already represent lines across the canvas. The process of reconstructing the image can be called a back projection. An intuitively simple way to perform the process is to add lines with varying darknesses based on the values on the sinogram. Unfortunately, it is not so simple. We illustrate why below:



(a) 2D demonstration overconcentration in the middle.



(b) 3D demonstration of overconcentration in the middle.

When we simply draw lines corresponding with each value on the sinogram, there is a large overlap in the centre of the reconstruction. The effects are blurred reconstructions that have no resemblance to the original image. The consequences are especially pronounced visually since the subject of most photos is placed in the centre of the image. As a result, we must take measures to mitigate the effect. However, the solution is quite convoluted. Which lines should we remove? How many should we remove? These questions will be answered with mathematical methods of the Fourier Transform.

4 Fourier Transform

To mitigate the effects of overlapping lines in the center of the reconstructed image caused by simple back-projection, we can utilize the Fourier transform. While delving into the deep mathematical derivations of the Fourier transform is unnecessary for our purposes, a basic understanding will guide us and explain its application in our string art construction. The algebraic expression of the transforms is also noteworthy. The solution to the string art relies on an elegant solution of rearranging the formulas.

4.1 One Dimensional Fourier Transform

The Fourier transform decomposes a signal into its constituent frequencies, representing it as a sum of sinusoidal waves. For example, the illustrations in Figure 5 demonstrate how a wave pattern can be expressed as the sum of simpler waves at different frequencies.

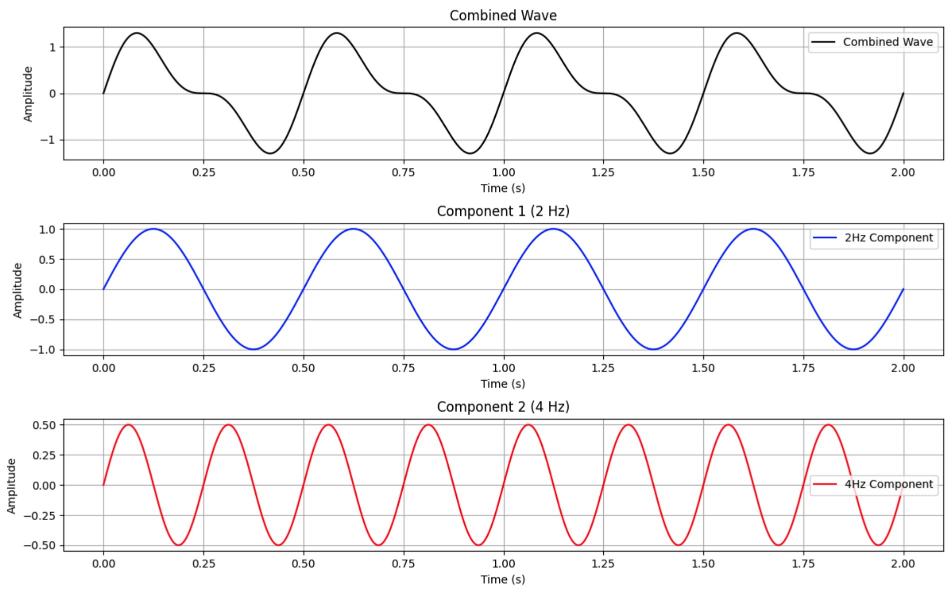


Figure 5: Decomposing a complex wave into two sinusoidal components

We can represent the waves as discrete values of frequencies. When we plot the discrete frequencies and their amplitudes in the combined wave, it is the expression of the wave in the **Fourier Domain**.

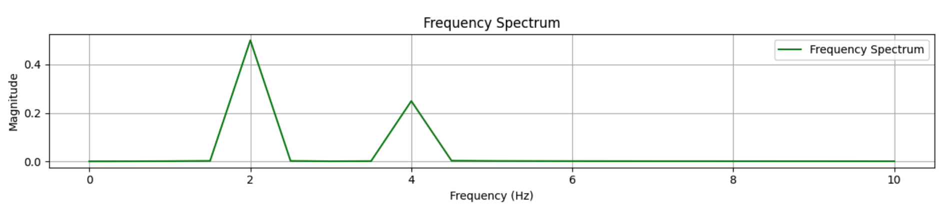


Figure 6: The frequencies of the components, also known as expression in the Fourier domain.

The mathematical proof of the Fourier transform is quite involved, and the derivation is not necessarily helpful to the understanding of our study. The basic understanding of the Fourier transform will suffice. However, the algebraic manipulation of the formula is still important. It is defined as follows.

Definition 1 *The Fourier transform of a one-dimensional function* $f(t)$ *is denoted as* $F_1\{f(t)\}$ *and is defined by:*

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt,$$

where ω *represents the angular frequency, and* j *is the imaginary unit.*

To convert the Fourier domain into the wave model, we can use the inverse Fourier transform, defined as follows.

Definition 2 *The inverse Fourier transform of a one-dimensional function* $F(\omega)$ *is denoted as* $\mathcal{F}^{-1}\{F(\omega)\}$ *and is defined by:*

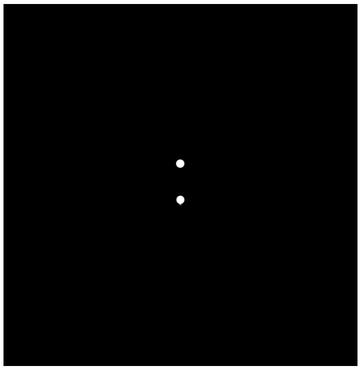
$$f(t) = \int_{-\infty}^{\infty} F(\omega)e^{j\omega t}d\omega,$$

where ω *represents the angular frequency,* t *represents time, and* j *is the imaginary unit.*

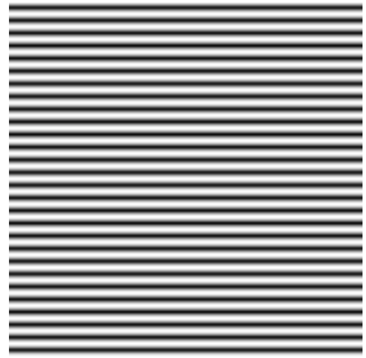
4.2 Two Dimensional Fourier Transform

Images are in two-dimensional. To work with them, we must extend the Fourier transform to the second dimension. The idea is similar. Additionally, I will provide an interesting example of the applications of the transform that will hopefully provide a better understanding of the transformations themselves and what the heatmaps all mean.

Firstly, let's explore the Fourier domain in the second dimension and the resulting images. The second-dimensional Fourier space is in two dimensions, which we can express as lightness in a 2d image.



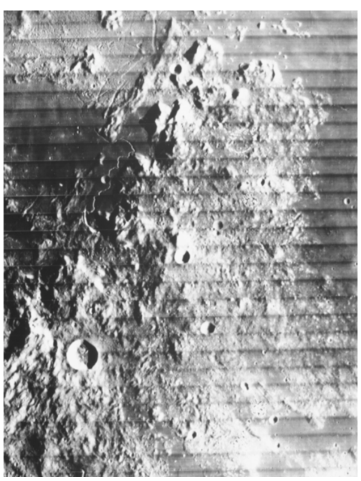
(a) Data in the 2D Fourier space.



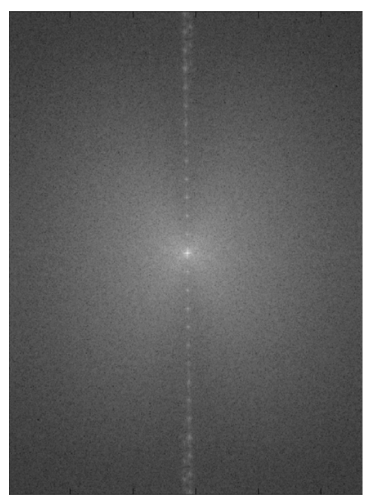
(b) The resulting projection from the dataset

As we can see, a point on the 2D Fourier space corresponds to a certain wavy pattern on the resulting projection. It is analogous to how the frequencies in the one-dimensional domain result in the oscillations. Most importantly, just as how any complex signal can be decomposed to its sine waves, any image can be represented by a pattern in the 2D Fourier space.

I will demonstrate an interesting use case to better illustrate the powerfulness of the 2D Fourier.



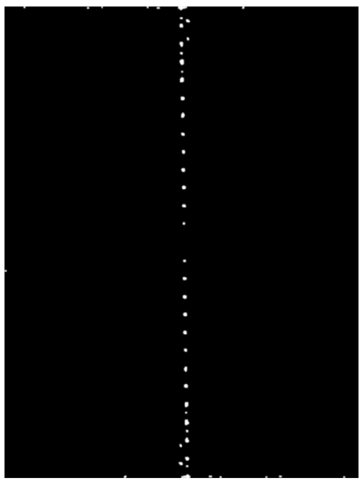
(a) Corrupted Moon Image



(b) The Fourier transform

In Figure 8a, we see an image of the moon that has streaks of lines running through it. In the corresponding 2D Fourier transform in Figure 8b, we can see bright spots that are consistent with the lines. We can isolate the peaks, shown in Figure 9a and remove them from the 2D Fourier space.

Then, if we project an image using the final 2D Fourier space with the peaks removed, we see that the interfering patterns are removed, as shown in Figure 9b.



(a) Peaks that are subtracted

(b) Uncorrupted Moon

We Mathematically express the definitions for these again, since they will become useful in the final derivation of the string art. Firstly, the forward 2D Fourier transform is the function that transforms an image into the 2D Fourier domain.

Definition 3 *The 2D Fourier transform of a two-dimensional function $f(x, y)$ is denoted as $F_2\{f(x, y)\}$ and is defined by:*

$$F_2(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy,$$

where u and v represent the angular frequencies in the x and y directions, respectively, and j is the imaginary unit.

The inverse, which is what allowed us to convert from the subtracted 2D space to the uncorrupted moon, will also be defined below.

Definition 4 *The inverse 2D Fourier transform of a two-dimensional function $F(u, v)$ is denoted as $F_2^{-1}\{F(u, v)\}$ and is defined by:*

$$F_2^{-1}\{F(u, v)\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv,$$

where u and v represent the angular frequencies in the x and y directions, respectively, and j is the imaginary unit.

4.3 The Fourier Slice Theorem

With an understanding of the Fourier domain, we can now examine the *Fourier Slice Theorem*, a powerful idea that links the Radon transform to the Fourier transform. The theorem states that the one-dimensional Fourier transform of a Radon projection of an image $f(x, y)$ at a given angle θ is equal to a slice of the two-dimensional Fourier transform of $f(x, y)$ along a line at the same angle θ through the origin.

Mathematically, this is expressed as:

$$F_1\{p(s, \theta)\} = F_2(u, v) \tag{4.1}$$

where $u = \omega \cos\theta$, $v = \omega \sin\theta$

5 Derivation of string art

We now have all the tools we need to derive the mathematical proof for the process of reconstructing images with string. The process is involved, and the summary of the steps will be covered in the next section.

Firstly, we start with the identity

$$f(x, y) = F_2^{-1} F_2(v_x, v_y) \tag{5.1}$$

We can expand the inverse 2D Fourier transform using Definition 4.

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F_2(v_x, v_y) e^{j2\pi(v_x x + v_y y)}\, dv_x\, dv_y.$$

We convert from Cartesian coordinates (v_x, v_y) to polar coordinates (ρ, θ), where:

$$\rho = \sqrt{v_x^2 + v_y^2}, \quad \theta = \arctan\left(\frac{v_y}{v_x}\right),$$

and the inverse relationships are:

$$v_x = \rho \cos\theta, \quad v_y = \rho \sin\theta.$$

The differential area element $dv_x dv_y$ transforms as:

$$dv_x dv_y = \rho \, d\rho \, d\theta.$$

After substitution, we obtain

$$f(x,y) = \int_0^{2\pi} d\theta \int_0^{\infty} d\rho \, \rho \, F(\rho\cos\theta, \rho\sin\theta) e^{j2\pi\rho(x\cos\theta + y\sin\theta)}$$

Here, from the Fourier slice theorem, $P(\rho, \theta) = F_2(\rho\cos\theta, \rho\sin\theta)$, we substitute and obtain:

$$f(x,y) = \int_0^{2\pi} d\theta \int_0^{\infty} d\rho \, \rho \, P(\rho,\theta) e^{j2\pi\rho(x\cos\theta + y\sin\theta)}$$

The significance of the substitution is we are able to express the object $f(x, y)$ with a Radon transform.

Additionally, since the radon transform is symmetrical past π, we can restrict the angular integration to $[0, \pi]$. Moreover, it means we can take the absolute value of ρ.

$$f(x,y) = \int_0^{\pi} d\theta \int_{-\infty}^{\infty} d\rho \, |\rho| \, P(\rho,\theta) e^{j2\pi\rho(x\cos\theta + y\sin\theta)}$$

We can also convert a part of the substitution back now, with $s = x\cos\theta + y\sin\theta$.

$$f(x,y) = \int_0^{\pi} d\theta \int_{-\infty}^{\infty} d\rho \, |\rho| \, P(\rho,\theta) e^{j2\pi\rho s}$$

Next, notice $\int_{-\infty}^{\infty} d\rho \;\, e^{j2\pi\rho s}$ is the inverse one dimensional Fourier Transform from Definition 2. Substituting, we obtain

$$f(x,y) = \int_0^{\pi} d\theta F^{-1}\{|\rho| P\{\rho, \theta\}\} \tag{5.2}$$

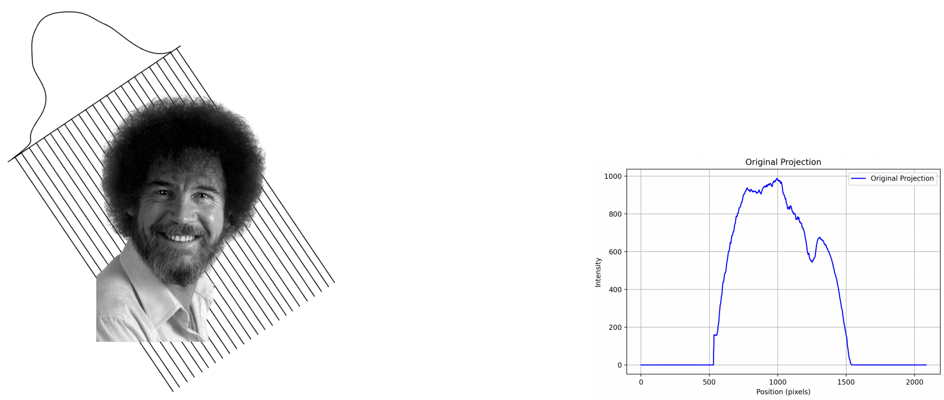
Lastly, we can expand out the function P, to express the process in terms of step-by-step processes stemming from the Radon transform.

$$f(x,y) = \int_0^{\pi} d\theta F^{-1}\{|\rho| F\{R\{s,\theta\}\}\} \tag{5.3}$$

Examining the expression, we see the function in the order as follows. Take the radon projection and transform it into the Fourier domain. Then, multiply by a filter, $|\rho|$ and take the inverse Fourier transform. Finally, integrate over all angles to obtain the original image. The idea is exactly how CT scanners work! Taking Radon transforms from all angles, and performing the process above to obtain an image of how the brain at that cross-sectional area looks like. The next section walks through the entire process.

5.1 The process

Begin with a radon transform of an image. We will use Bob Ross.



(a) Radon transform

(b) Value of the transformation

Next, we perform the Fourier transform the projection to obtain the transform in the Fourier domain.

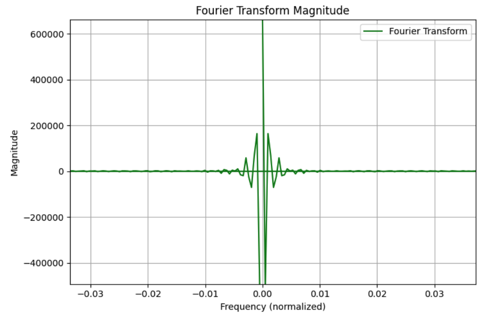
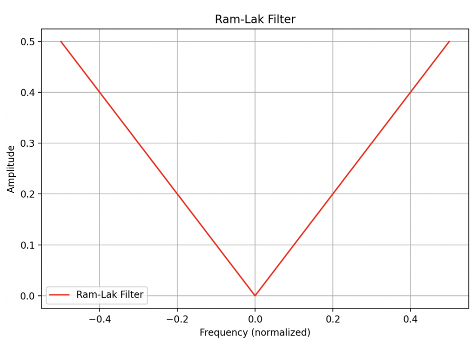
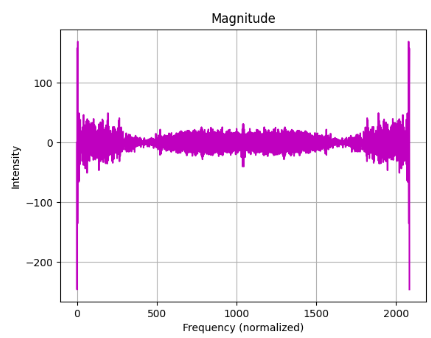


Figure 11: Transformation of projection to the Fourier domain

Now, we can clearly see the problem. The low frequency values near zero are far too prominent, which lead to the overconcentration that was outlined in the beginning of the paper! To remove them, we can multiply by the filter, $|\rho|$. Figure 12a demonstrates what the filter looks like. Notice how the function will suppress the overconcentrated low frequencies and allow the distinct, higher frequencies to pop out.



(a) The high pass filter, $|\rho|$



(b) After multiplication with the filter.

We perform the inverse Fourier on the frequencies to obtain the intensities of light at each interval of the projection. The end result we obtain is this.

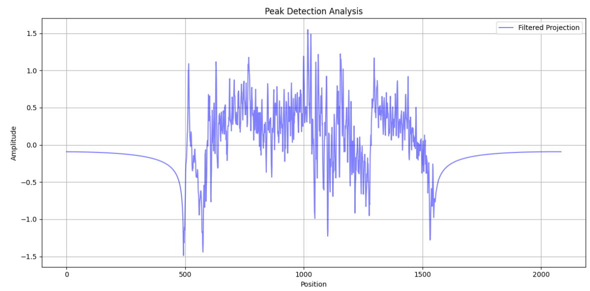


Figure 13: The Fourier domain to the pixel intensities

Now if we extend the one dimensional intensity into an actual layered projection, we can get a smeared-looking view.

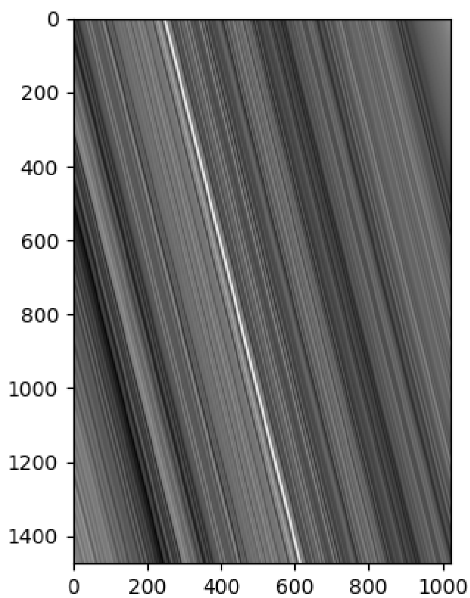


Figure 14: Smeared values of the one-dimensional pixel intensities

Finally, for the step of integrating all projection over all angles, we will stack each projection and obtain a reconstruction. Through summing smeared one dimensional projections, we can use the process we just outlined to reconstruct the images without loss of information.

Figure 15: Reconstruction of Bob Ross with smeared projections

6 Discritization of Lines

Now, we begin to deviate from the medical field in pursuit of art. For string art, all processes up to the filtering remain the same. However, now, instead of creating a smear image, we create discrete lines. Recall Figure 13. Last time, we smeared the projection based on their intensities. Now, to create string art, we can simply place string at the highest peaks. Examine Figure 13. We find the ten highest peaks since they represent the places of the greatest darkness. It is at these pixel values that we should draw straight lines across the canvas. Figure 16 below demonstrates the peaks.

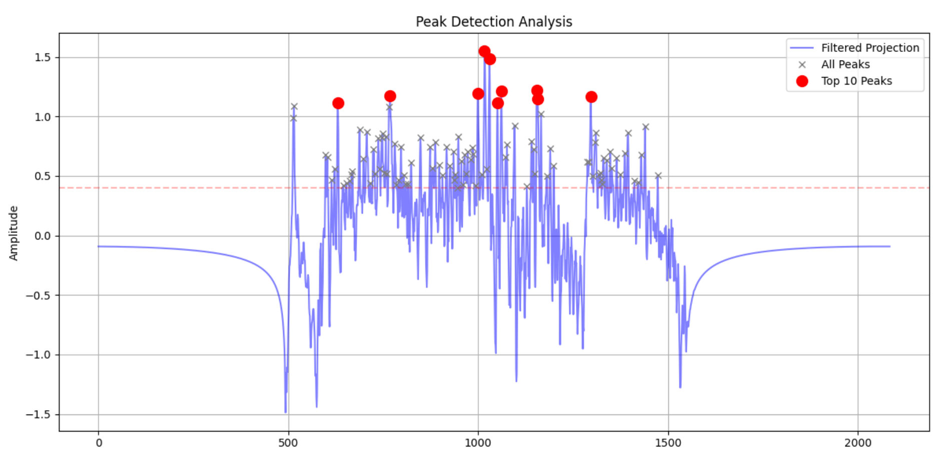


Figure 16: Peaks in the pixel intensities

We can draw lines at these peaks, and stack them in a similar manner as the previous reconstruction. The only difference this time is that the final result looks exactly like string art.

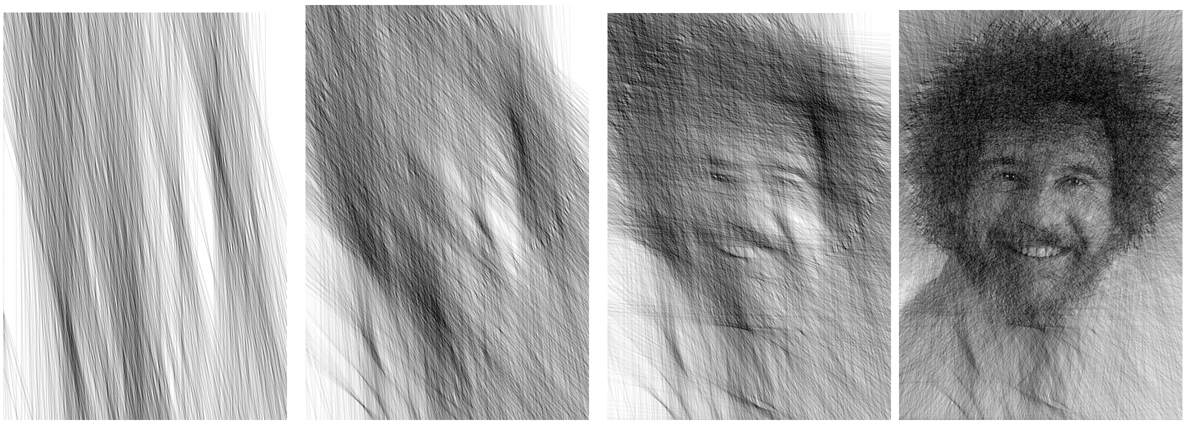


Figure 17: Reconstruction of Bob Ross with string.

7 Improvements

We now have a model that can create string art to a recognizable extent. The bulk of the methodology comes from the mathematical derivation to the equations of Equation 5.3. However, it is by no means is the model the best it could be. There are many other parameters not included in the process that may influence the resulting image. This section outlines the adjustable factors that we can change to seek to improve the final result.

7.1 Image pre-treatment

Firstly, I found that some images work better than others. Specifically, when there is a subject in the centre of the image, the reconstruction seems to work much better. Additionally, transparent or white backgrounds interfere with the Radon transform much less. Consequently, it is advised to remove any background and centre the subject prior to processing it through the algorithm.

7.2 Number of peaks

Choosing the number of peaks that correspond to a string also affects the final result. Too much string can cause overlap which leads to dark images with no detail. Too little string can lead to a loss of detail in the final projection. After playing around for a bit, I found that 10 peaks, or strings, per angular projection yielded the best results.

8 Gallery

With the improvements made, here are some of the best results I was able to produce with the algorithm.

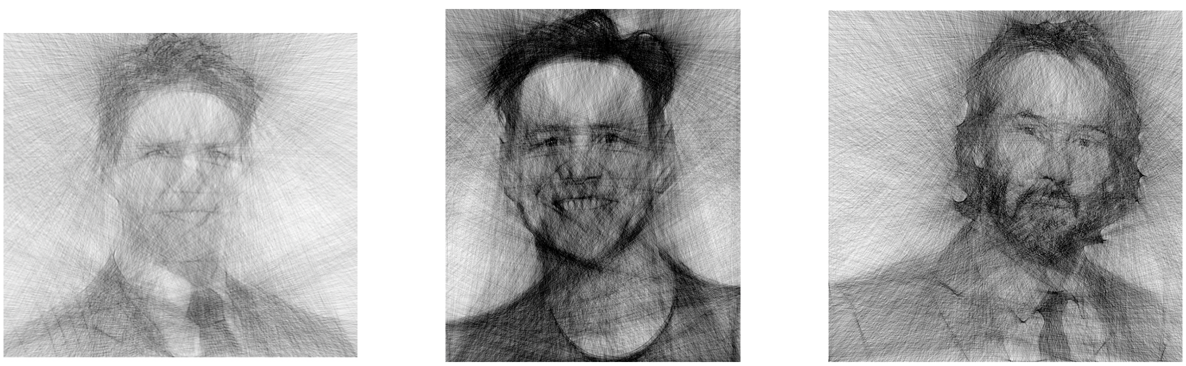


Figure 18: Gallery of celebrities drawn with string art.

9 Conclusion

In this paper, we developed a mathematical framework for creating string art by leveraging concepts from Radon transforms, Fourier analysis, and back projection techniques. Through a detailed derivation and exploration of the Fourier Slice Theorem, we demonstrated how methods originally designed for medical imaging can be adapted for artistic purposes. The results highlight the potential of mathematical tools to inform and inspire creative expression. While the current implementation achieves visually striking results, opportunities for improvement—such as optimizing image preprocessing and adjusting line selection parameters—offer exciting avenues for future exploration. Ultimately, this work illustrates the synergy between mathematics and art, showcasing how analytical rigour can lead to innovative and beautiful creations.

I know this is still a draft, but what I eventually want to do is actually create a physical model of the string art, I was wondering if thats okay to include here too?